

# AKHILENDRA DWIVEDI

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## PROFESSIONAL SUMMARY

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M.Tech AI Candidate at NIT Jalandhar with a background in full-stack web application development, microservices, and Retrieval-Augmented Generation (RAG) pipelines. Currently conducting academic research in Wi-Fi Sensing, processing high-dimensional signal analytics and sensor datasets for machine learning applications.

## EDUCATION

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| <b>Master of Technology in Artificial Intelligence</b>                   | Aug 2025 - June 2027 |
| Dr B R Ambedkar National Institute of Technology (NIT) Jalandhar, Punjab | CGPA: 8.35/10.00     |

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| <b>Bachelor of Technology in Information Technology</b>                | Nov 2020 - June 2024 |
| Dr Ambedkar Institute of Technology for Handicapped (AKTU), Kanpur, UP | CGPA: 8.06/10.00     |

## SKILLS

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**Languages:** Python, JavaScript, SQL, C++, C, HTML/CSS  
**AI & Machine Learning:** Deep Learning, NLP, LLM Fine-Tuning, Embeddings, Scikit-Learn  
**Agentic Pipelines:** RAG, Multi-Agent Frameworks, Knowledge Graphs, LangChain  
**Databases:** Neo4j, Qdrant, ChromaDB, MongoDB Atlas, MySQL  
**Backend & Frameworks:** FastAPI, Node.js, Express.js, React.js, Vite, Tailwind CSS  
**Developer Tools & Core CS:** Docker, Docker Compose, Git, Data Structures & Algorithms, OOPs

## PROJECTS

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| <b>Gov Scheme AI – AI-Powered Recommendation Platform</b> | June 2026 - Present |
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- Engineered a hybrid retrieval system matching public policies via dense **semantic search (Qdrant)** and rule-based filters.
- Built high-performance asynchronous microservices using **FastAPI** and **Node.js/Express** to handle natural language inputs.
- Integrated **OpenRouter model** for multilingual generation and containerized the architecture using **Docker** and **Docker Compose**.

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| <b>Narrative Consistency Detection, KDSH Finalist, IIT Kharagpur x Pathway</b> | Dec 2025 - Jan 2026 |
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- Architected a **RAG** reasoning pipeline utilizing **Pathway ETL** for stream data processing and text anomaly detection.
- Deployed open-weight LLMs (**Qwen2.5-7B**) locally via **Ollama** to run complex contextual logic verification on text datasets.
- Utilized **Sentence Transformers (all-mpnet-base-v2)**, **Scikit-Learn**, and **Pandas** to compute semantic similarity thresholds.

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| <b>LitNavigator AI – Multi-Agent Research Intelligence Platform</b> | March 2026 - April 2026 |
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- Built an autonomous research engine using **LangChain** multi-agent architectures to orchestrate recursive literature reviews.
- Designed a hybrid retrieval framework leveraging **ChromaDB** for dense vector lookups and **BM25** for sparse keyword matching.
- Constructed **Knowledge Graphs** in **Neo4j** to map citation dependencies, evaluating agent streams with **LangSmith**.